

Mann §3.1.2 — Representations

A Plain-Language Walkthrough (with diagrams)

Reading companion — Mann, *An Introduction to Particle Physics and the Standard Model*, Ch. 3

2026-05-18

Where we are. In §3.1.1 we learned that a *symmetry* is described by a *group*: a set of transformations that is closed, has an identity, has inverses, and is associative. That was abstract — just symbols g_i and a rule “ $\bullet$ ”. Section 3.1.2 answers the practical question: *what do these transformations actually act on, and how do we compute with them?* The answer is **matrices**, and the bridge is the idea of a **representation**.

1. What does a symmetry transformation act *on*?

In particle physics, the object that gets transformed by a symmetry is almost always the **wavefunction** of a particle (or set of particles). Mann points out that, unlike intro QM where the wavefunction is a single complex function, here it is generally a **multi-component complex column vector** living in some abstract space. A symmetry operation acts on it by **matrix multiplication**.

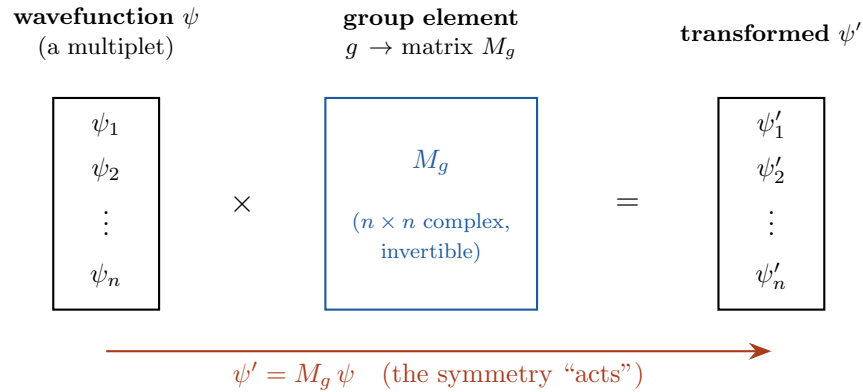


Figure 1: A symmetry transformation g is realized as an $n \times n$ complex matrix M_g that multiplies the wavefunction column vector ψ and “rotates” it into a new one ψ' . The number of components n is the *dimension* of the representation.

Reading the figure. The left box is the particle’s wavefunction written as a stack of complex numbers (a *multiplet*). The middle box is the abstract group element g *turned into* a concrete matrix M_g . Multiplying gives a new wavefunction ψ' . Crucially, the matrix must be *square and invertible* — those requirements are forced on us by the group axioms, as we will see.

2. The core idea: a representation is a “translation dictionary”

A group is abstract; matrices are concrete and computable. A **representation** is a rule that assigns a matrix M_g to every group element g in a way that respects the group’s multiplication table:

Definition (in words and symbols)

For every $g \in G$ there is a matrix M_g such that

$$g_1 \bullet g_2 = g_3 \implies M_{g_1} M_{g_2} = M_{g_3}.$$

“Combine first, then translate” gives the *same* answer as “translate first, then multiply the matrices.” Such a structure-preserving map $\rho : G \rightarrow \{M_g\}$ is called a **representation** (mathematically, a *homomorphism*). A d -dimensional representation uses $d \times d$ matrices.

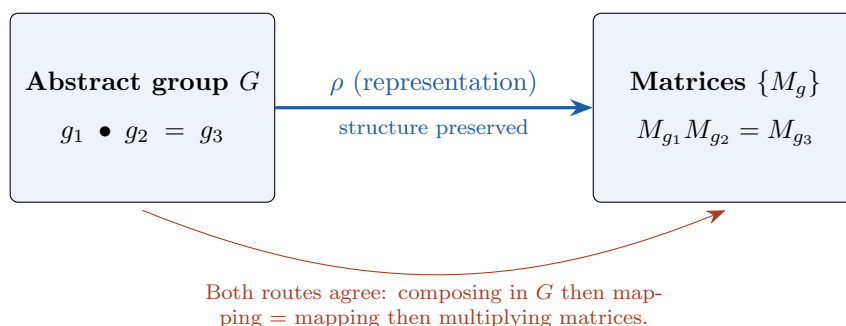


Figure 2: A representation ρ is a “dictionary” from abstract group elements to matrices that keeps the multiplication table intact. This is the single most important sentence in §3.1.2.

Why this is powerful. Once the dictionary exists, every abstract symmetry question becomes ordinary (if tedious) **matrix algebra** — something we can actually compute.

3. Why matrices are the perfect choice

Mann notes three “freebies” you get automatically by using matrices:

- **Associativity is automatic** — matrix multiplication is associative, so that group axiom is satisfied for free.
- **Inverses force invertibility** — the group’s inverse axiom means every M_g must be an *invertible* matrix. This sharply restricts which matrices can appear.
- **Identity is trivial** — the group identity I is just the identity matrix $\mathbb{I}$.

4. Three things that surprise people

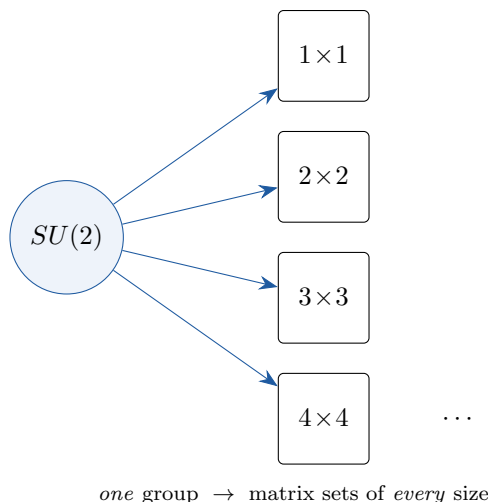


Figure 3: (a) **Representations are not unique.** The very same group — here $SU(2)$ — can be represented by matrices of dimension $1, 2, 3, 4, \dots$. A group has *infinitely many* representations.

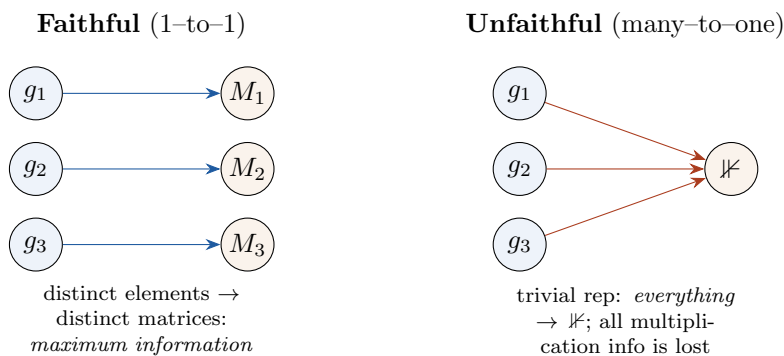


Figure 4: (b) **Representations need not be faithful.** If the map is 1-to-1 (left), the representation is *faithful* and carries the maximum information about the group. Often several elements share one matrix (right); the extreme case maps *everything* to the identity.

And the third point:

- **The fundamental representation.** If the group already *is* a set of matrices, that set is automatically a faithful representation of itself — called the **fundamental representation**. (Example: $SU(2)$ as 2×2 unitary matrices is its own fundamental rep, but it *also* has the $1, 3, 4, \dots$ dimensional reps from Fig. 3a.)

5. Why ever use an *unfaithful* representation?

It seems wasteful to throw away information — so why bother? Mann's footnote gives the key example: map every group element to $\pm \mathbb{I}$. You lose almost everything, but you *keep* one crucial

bit: the even/odd character of each element. For the rotation group this single bit tells you whether a transformation is a **reflection**, i.e. whether it has **odd parity** — already real physics. Lesson: use the smallest representation that still carries the information you actually need.

Common confusions to avoid

- “Representation = the group.” **No** — it is one *matrix picture* of the group; there are infinitely many.
- “Bigger matrices = more fundamental.” **No** — the *fundamental* rep is the group’s own defining matrices; larger reps are derived.
- “Unfaithful reps are useless.” **No** — they isolate a specific feature (e.g. parity) cleanly.
- Dimension of the rep = number of wavefunction components = size of the multiplet (doublet, triplet, ...).

One-paragraph takeaway

A **representation** turns the abstract group of a symmetry into concrete $d \times d$ invertible complex matrices that act on a particle’s wavefunction multiplet, *preserving the group’s multiplication table*. The matrix viewpoint makes associativity, inverses, and the identity automatic and lets you actually *calculate*. A group has infinitely many representations of many dimensions; they may be *faithful* (1-to-1, maximal information) or not; the group’s own defining matrices form the *fundamental* representation. Choosing a representation is choosing how much of the symmetry’s information you want to keep for the physics at hand — which sets up the next section’s question: what are the *irreducible* (non-redundant) representations?